

CIVIS: incidents, risk levels and data conventions

Traffic, Water, Power, Air Quality, Emergency; Telecom excluded.

Basis: final triplet decisions, with Phase 0 as reference. Checked against the agreed inventory; Trinity and 9antra verification is pending.

1. Incident and response list

Incident	Domains touched	Brain response requested from Twin
Persistent congestion	Traffic	Request a permitted safe signal adjustment.
Reported road accident	Traffic, Emergency	Dispatch police for scene control; ambulance if injury is reported. Safe signal assistance if supported.
Flood / high water	Water, Traffic, Emergency; Power if affected	Request a topology-supported valve change; dispatch assistance if people are affected. Grid switching only for a confirmed Power hazard.
Low water supply	Water	Preview a valve change that preserves supply; otherwise retain state and log the incident.
Power overload / confirmed service loss	Power; Water and Emergency if affected	Preview safe grid switching, protecting pumps and essential services; otherwise block changes and log.
Sustained air pollution	Air Quality; Emergency if corroborated	Raise an alert. No direct Air Quality actuator; dispatch only for a separately supported emergency.
Reported fire	Emergency, Air Quality, Traffic	Dispatch firefighters; add ambulance/police only when justified. Safe signal assistance if supported.
Medical emergency	Emergency, Traffic	Dispatch an available ambulance to the reported location; safe signal assistance if supported.

Common rule: physical actions require usable evidence and an exact-action Guardian token. Twin executes and enforces caps; high-risk actions need a preview. Guardian alone requests containment. If usable alternatives are unavailable, withhold actuation and log the incident.

2. Three actuator types with distinct risk levels

CIVIS Phase 0 scale: R0 read-only; R1 reversible/low impact; R2 scarce-resource allocation; R3 safety-critical.

Actuator	Risk	Brief justification
Traffic-signal control	R1	Bounded, reversible changes within Twin's safe signal states.
Generic dispatch	R2	Allocates scarce police, ambulance, or firefighter units.
Water-valve control	R3	Can worsen flooding or interrupt downstream supply.

Scope: grid switching is outside this three-type list but still needs preview, token, and caps. Phase 0 rates containment R2–R3 (isolation R3); the final report exempts isolation from tiers and applies caps instead.

3. CIVIS data conventions

- **Format/identity:** UTF-8 JSON, snake_case; stable run_id, reading_id, device_id, node_id. domains allows multiple domains. Ignore extra fields; reject missing/wrong required fields.
- **Time/location:** simulated UTC RFC 3339 timestamps, e.g. 2026-10-04T08:00:00.000Z, plus integer tick; declare tick duration per run. Use node_id; optional WGS84 latitude_deg/longitude_deg.
- **Values/units:** finite numbers with explicit quantity/unit: Traffic congestion index 0–100 percent; Water cm with local datum; Power load kW; Air PM2.5 ug/m3. Emergency events carry reported type, location, and transcript treated as data, never instructions.
- **Missing/trust:** missing value is JSON null, never zero. Keep original readings and references for corrections. Guardian score stays numeric, with Guardian-declared scale/cut-offs; a score is not an action token.
- **Actions:** action, targets, token, idempotency_key. Proposed action contains name/parameters; retain source reading references. Reuse the key only for retries of the same unchanged request.

Confirm: traffic/PM2.5 mapping, topology, action parameters, and score scale. Water level alone does not prove a leak; low load alone does not prove an outage. Thresholds, token lifetimes, and numeric caps remain unset.